AI-led Post-Event Content Automation: Working Prototype

A live build that takes a webinar's registrant list and recording and turns them, with AI as the engine, into enriched and routed leads, segment follow-up emails, a multi-channel content package, and a lead intelligence dashboard.

Rahul Ranjan · 13 October 2026

Links

Content package (Google Drive)	https://drive.google.com/drive/folders/1p9J4D3ZeTDCJnUkRY4GSgCAYoT3ZZ9EB
Code repository	https://github.com/irahulstomar/darwinbox-post-event-ai

The build runs on my self-hosted n8n. I would be happy to walk you through exactly how it works, live, over a call.

Results at a glance

- Module 1: contact and company data completeness went from 42% to 99%; 68 rows de-duplicated to 64 people, 13 merged into existing HubSpot records.
- Module 2: three segment emails sent through HubSpot within minutes of the event closing, with UTM tracking.
- Module 3: full content package generated; 6 of 7 AI YouTube chapters match the creator's official chapters.
- Module 4: live dashboard with an AI narrative, AI-adjusted interest scores and anomaly detection that found all three planted patterns.

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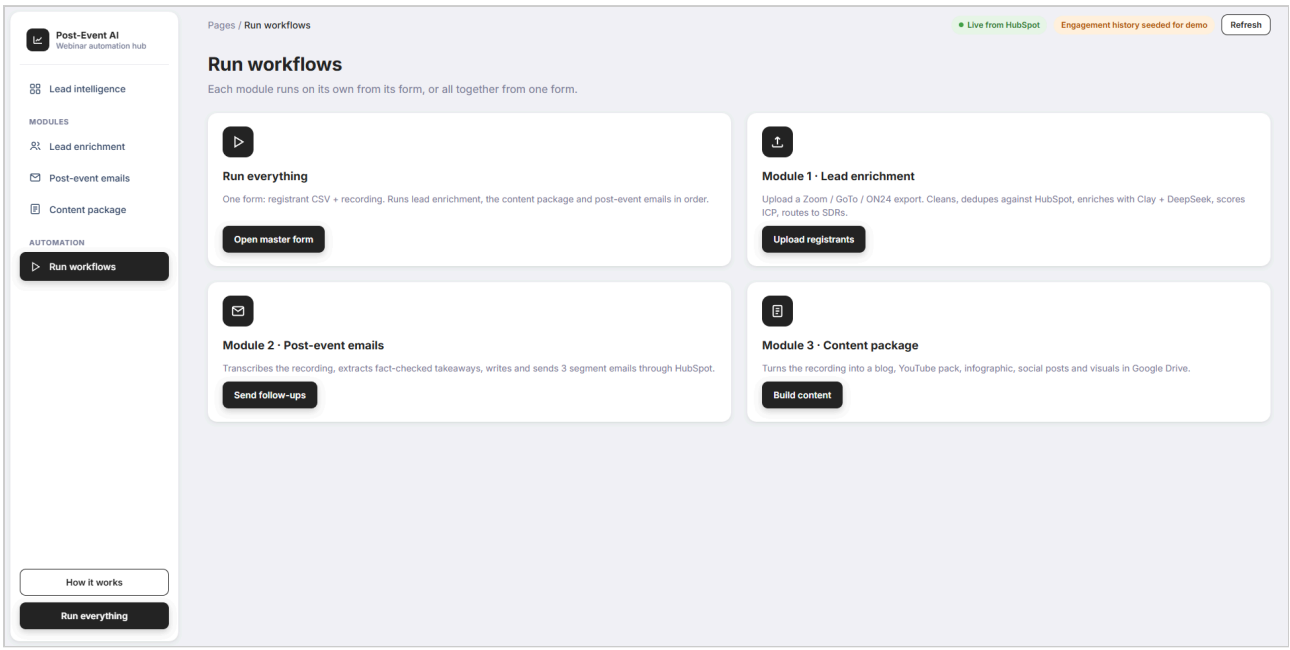
1. How it works: architecture and stack
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1. Architecture and stack

Every module is its own n8n workflow with its own form, so each can be demoed alone. A control hub (served by n8n) shows the results of all four, live from HubSpot and Google Drive, and a master workflow can run M1, M3 and M2 in order from one form. The only human step is the trigger (upload a list or a recording link), which stands in for the webinar platform's webhook. Every decision after that is made by AI and checked by code.

1. Inputs: registrant export (Zoom format), recording or transcript, watch and CTA links.
2. M1 Enrich: clean, dedupe against HubSpot, Clay, DeepSeek, ICP score, route.
3. M2 Email: Deepgram transcript, takeaways, fact guard, three segment emails through HubSpot.
4. M3 Content: blog, YouTube pack, infographic, social posts, Gemini visuals, Google Drive.
5. M4 Intelligence: live dashboard with anomalies, interest scores and an AI narrative.

Layer	Tool	Used for
Orchestration	n8n (self-hosted, 2.12)	All workflows, forms, webhooks, the hub page and its data endpoints
CRM	HubSpot (developer test account, Enterprise trial)	Dedupe, property mapping, owners, lifecycle stages, lists, marketing email send and tracking
Enrichment	Clay (webhook table + HTTP callback)	Company domain lookup and company enrichment (size, industry, LinkedIn, description)
LLM	DeepSeek V4 (flash for classification, pro for writing and analysis)	Normalisation and inference, ICP reasoning, takeaways, email copy, content, anomaly judgement, narrative
Transcription	Deepgram nova-3 (diarization + keyterm prompting)	Recording to timestamped, speaker-labelled transcript
Image generation	Gemini 3 Pro Image (Google Cloud)	YouTube thumbnail with title text, infographic poster, LinkedIn visual
Storage	Google Drive (Docs API for formatting)	Content package per event, formatted Google Docs, README with QA checks
Dashboard	HTML + Chart.js, served by n8n	One URL for all modules; API tokens never reach the browser



Control hub, "Run workflows" page: each module runs from its own form, or all together from the master form.

2. Demo data: what is real and what is synthetic

Darwinbox's real registrant lists and engagement data are private, so the demo combines a real webinar with a realistic synthetic audience. Nothing here can reach a real person.

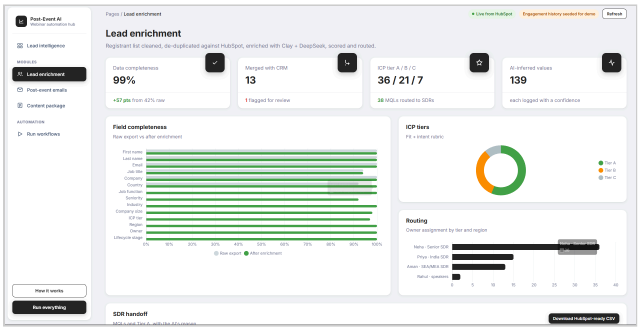
Input	Source	Notes
Webinar recording	Real, public	"Darwinbox Cortex and the AI-Native Era of HCM": OutSail (Brett Ungashick) in conversation with Jayant Paleti, co-founder of Darwinbox. 37 min, transcribed with Deepgram.
Registrant export	Synthetic, modelled on a Zoom attendee report	68 rows: 45 attended, 21 no-shows, 2 panelists, 4 duplicate sign-ups. Real companies (India, SEA, Middle East, US), fake people, deliberately messy (Gmail sign-ups, blank titles, "TATA STEEL" vs "Tata Steel Limited", swapped name order).
Existing CRM	Synthetic	14 contacts pre-loaded in HubSpot with different spellings and emails, so dedupe has real work to do.
Email addresses	Safe by design	All demo contacts use the reserved `.example.com` domain (no mail can be delivered). Speakers and two demo inboxes are aliases of my own inbox.
Engagement history (M4)	Seeded, labelled in the dashboard	The event was yesterday, so 30 days of opens, clicks, page views, form fills and stage history were seeded, including 3 planted patterns for the AI to find. Real HubSpot opens from the sends are shown alongside.

HubSpot test-account rule. Developer test accounts only deliver marketing email to users of the account. The real sends to the speakers and demo inboxes were delivered (and opened); sends to the synthetic contacts are logged in HubSpot as dropped. A paid Marketing Hub account delivers to everyone with no change to the workflow.

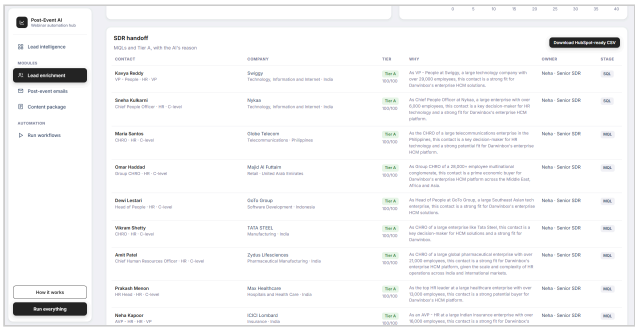
3. Module 1: Lead list enrichment

Upload the registrant export to the M1 form. Three n8n lanes do the rest: intake (clean, dedupe, save to HubSpot, send to Clay), enrichment (Clay calls back per row, DeepSeek classifies, code scores and routes) and a live report.

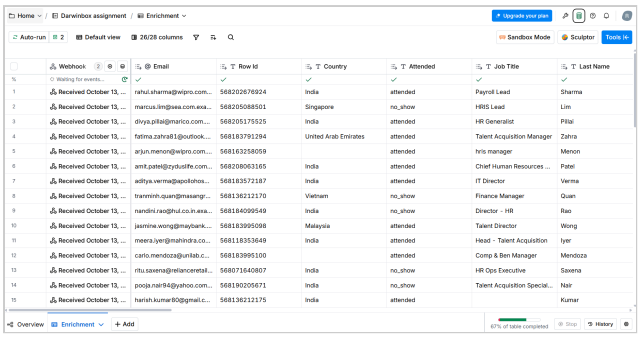
Brief asks for	What the build does
Fuzzy match and dedupe against HubSpot	In-file dedupe (same email, or same person at the same company via Jaro-Winkler + company token match, incl. swapped name order), then HubSpot matching: exact email (1.0), same name + company (0.95, merge), similar name + company (0.8, flagged for review). Every decision is logged with its reason. Result: 68 → 64 people, 13 merged, 1 "Priya vs Priyanka Das" flagged, two different "Rahul Sharma"s kept apart.
Infer missing title, function, seniority, size, industry	Clay first (company size, industry, HQ, LinkedIn); DeepSeek fills what is missing and records each inferred value with a confidence and basis (139 inferred values across the list). Country is inferred from the phone code or company HQ.
Enrich company data, score ICP fit	Editable rubric (size, region, HR-buyer persona, seniority, attendance and engagement) plus a one-line AI reason. Tier A needs an HR-buyer persona; students, consultants and vendors are capped.
HubSpot-ready output, completeness above 90%	Completeness 42% → 99% across 14 SDR fields. Region assigned, owner routed (Tier A to senior SDR, others by region), lifecycle set forward-only (attended + Tier A/B + known role = MQL). CSV download in the hub.



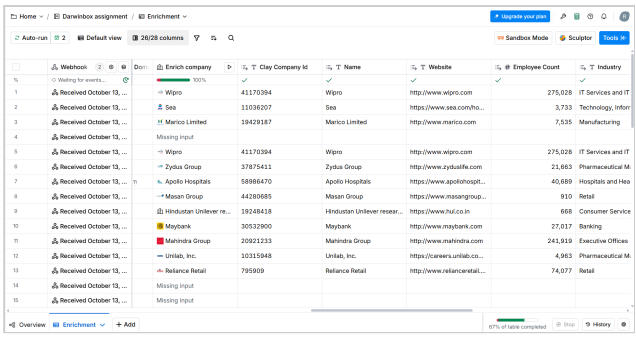
Hub · Lead enrichment: completeness before vs after, tiers, routing.



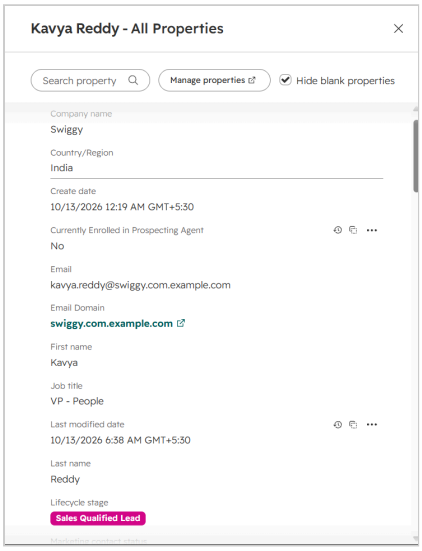
SDR handoff with tier, score, AI reason, owner and stage.



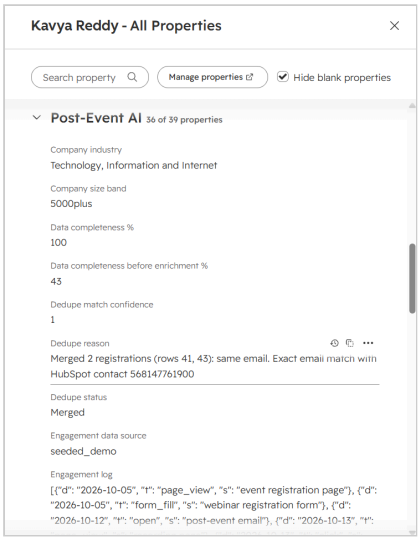
Clay table receiving rows from n8n (webhook source).



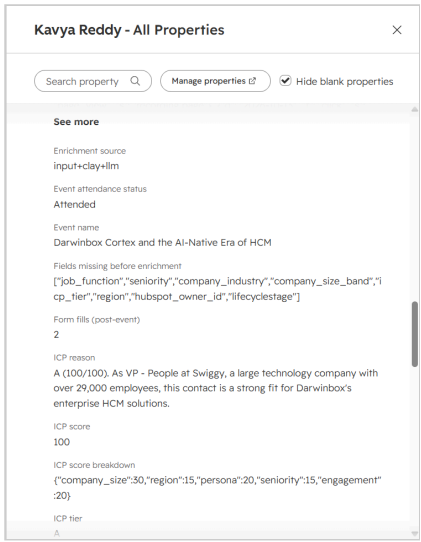
Clay enrichment columns: domain lookup, company enrichment, employee count, industry.



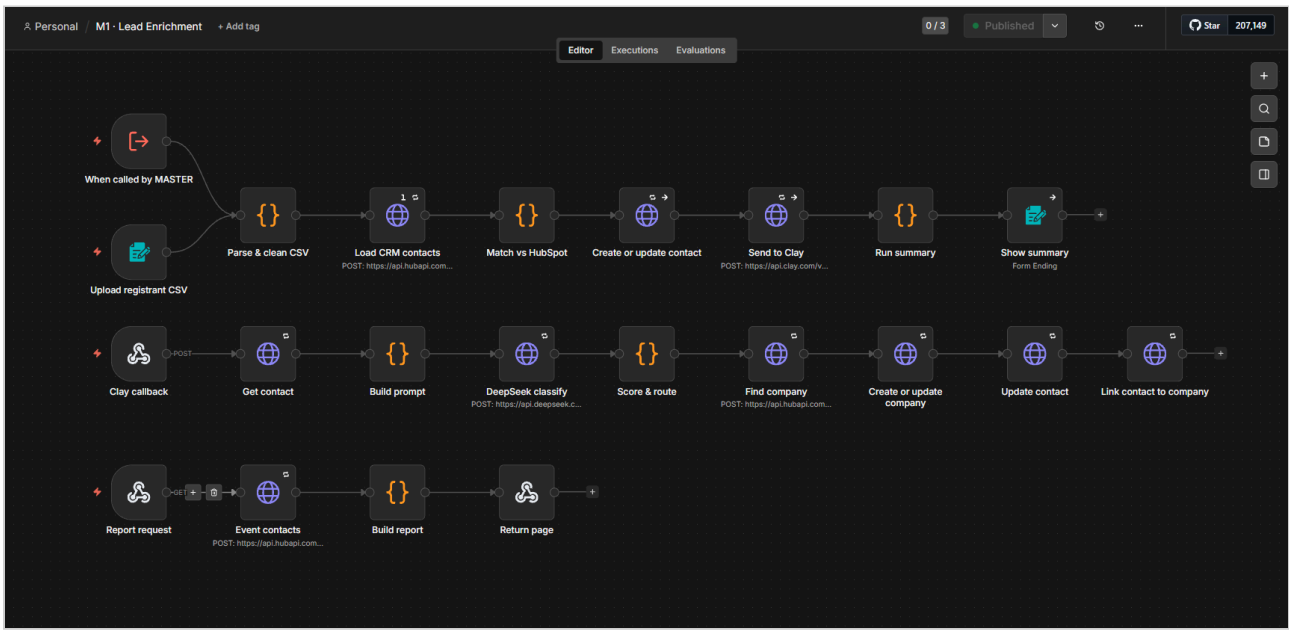
HubSpot contact (Kavya Reddy): standard fields updated.



"Post-Event AI" property group: completeness, dedupe reason, engagement.



ICP score, breakdown and AI reason.

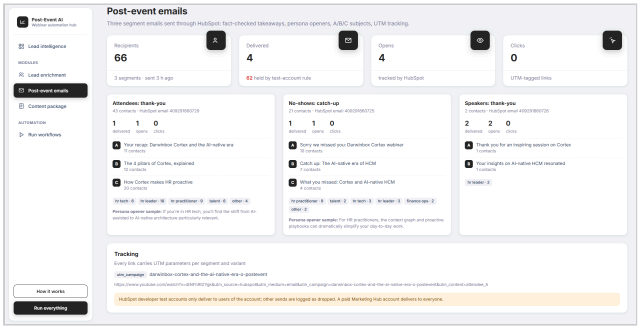


n8n · M1 workflow: intake lane (form or master trigger), Clay callback lane, report lane.

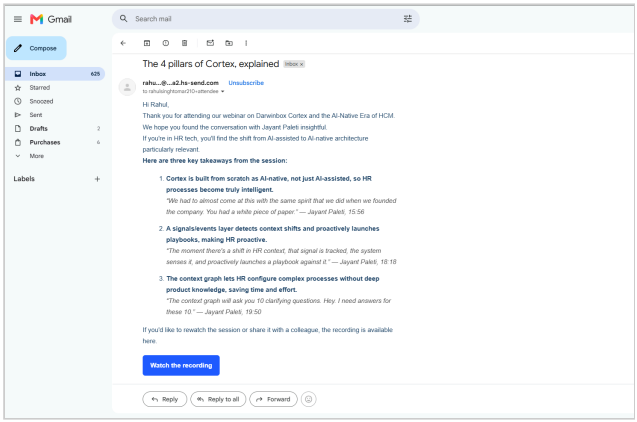
4. Module 2: Post-event communications

The "Event closed" form takes the recording link (or a transcript as a fallback). Deepgram transcribes it with speaker labels, DeepSeek extracts takeaways with verbatim quotes, code fact-checks them, DeepSeek writes the copy, and HubSpot sends and tracks three segment emails.

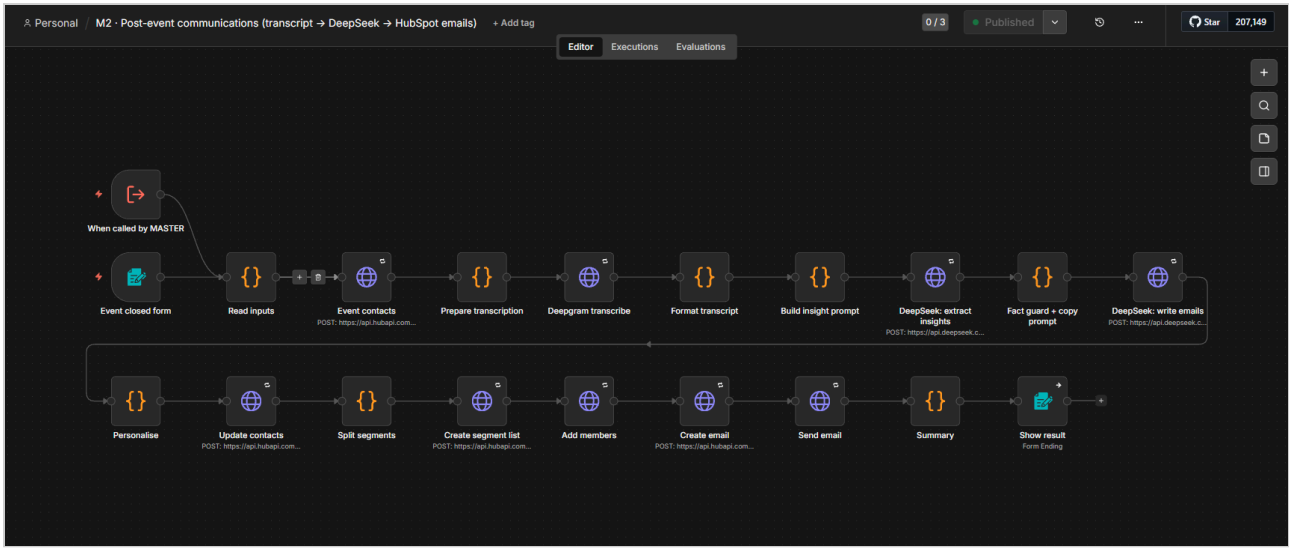
Brief asks for	What the build does
Segment-specific copy and subject variants	Attendee thank-you (recording + 3 takeaways), no-show catch-up (recording first + takeaways), speaker thank-you (performance snapshot). Three subject lines per segment, assigned A/B/C per contact.
Short key-takeaway snippets	Three "so what for an HR leader" snippets, each paired with a verbatim, timestamped quote that passed the fact guard.
Personalise by role or industry	A persona opener (HR leader, HR practitioner, HR tech, talent, finance/ops) and an industry line, written into each contact's HubSpot record and merged into the email with personalisation tokens.
Speaker performance snapshot	Computed by code, never generated: registrants, attendance rate, average minutes, % staying 30+ min, questions, % Director+, top industries, plus the speaker's best verified quote.
Dispatched within 24 hours, logged with UTMs	Sent through HubSpot marketing email within about a minute of the form. Segment lists and emails are created per run; every link carries utm_source, utm_medium, utm_campaign and utm_content = segment + variant. Sends, opens and clicks are tracked by HubSpot.



Hub · Post-event emails: per-segment delivery, opens, A/B/C subjects, persona mix.



A delivered email: personalised greeting and opener, fact-checked takeaways with timestamps, tracked recording link.



n8n · M2 workflow.

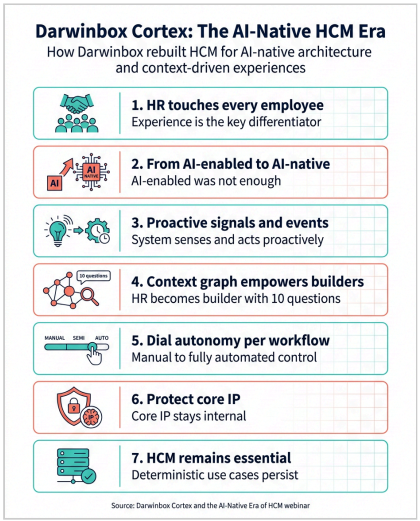
5. Module 3: Content repurposing

From the recording, M3 extracts key moments, insights, verbatim quotes and data points, fact-checks them, then generates each asset, validates it against hard rules in code, creates the visuals, and saves everything to a per-event Google Drive folder as formatted Google Docs, tagged by event, with a README index and QA report.

Brief asks for	What the build does
Blog draft, 800-1200 words	Generated as structured parts and assembled by code; quotes are inserted from the verified list (guaranteed word-for-word); length enforced by code (expand pass if short, trim whole secondary paragraphs if long).
YouTube chapters, description, thumbnail brief	Chapters validated against YouTube's rules (start 00:00, 3+, 10s apart). Compared with the creator's official chapters: 6 of 7 within 90 seconds. Description + tags + thumbnail brief, and a finished thumbnail image.
Infographic outline with data points	5-7 sections built only from fact-checked data points and quotes, plus a data-accurate HTML version and a rendered poster.
5-10 social posts with distinct hooks	5 LinkedIn + 5 X posts; one idea per post; hook types distinct per platform; X length checked.
Image generation, saved to a shared drive, tagged	Gemini 3 Pro Image in two house styles chosen from a style test; images throttled to respect quota with automatic retries; a note is saved instead if an image fails, so the run still completes.



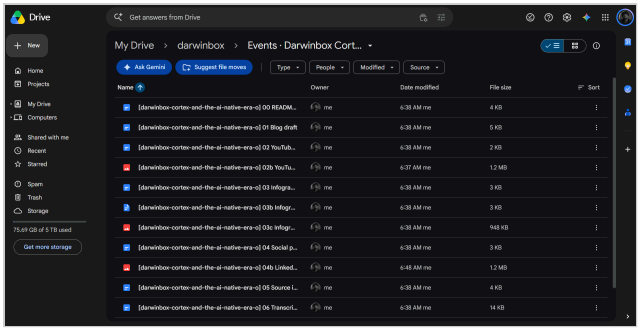
YouTube thumbnail (AI, with title text).



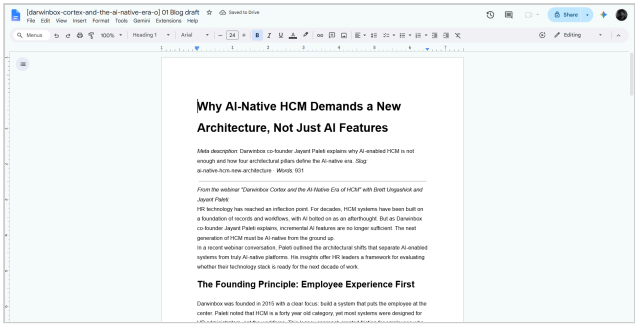
Infographic poster (AI, fact-checked text).



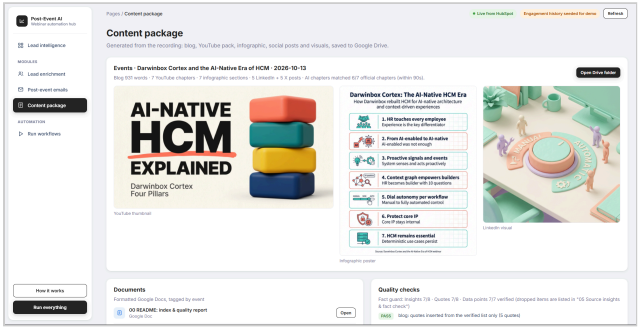
LinkedIn visual (AI).



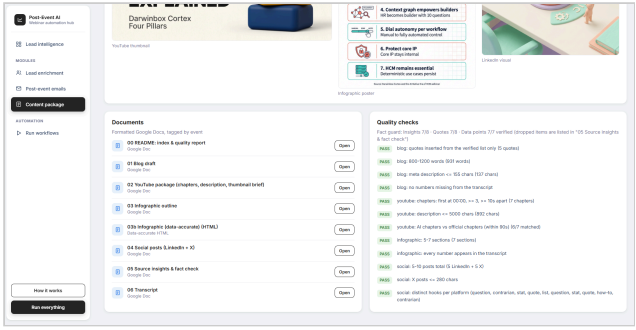
Google Drive: one folder per event, every file tagged.



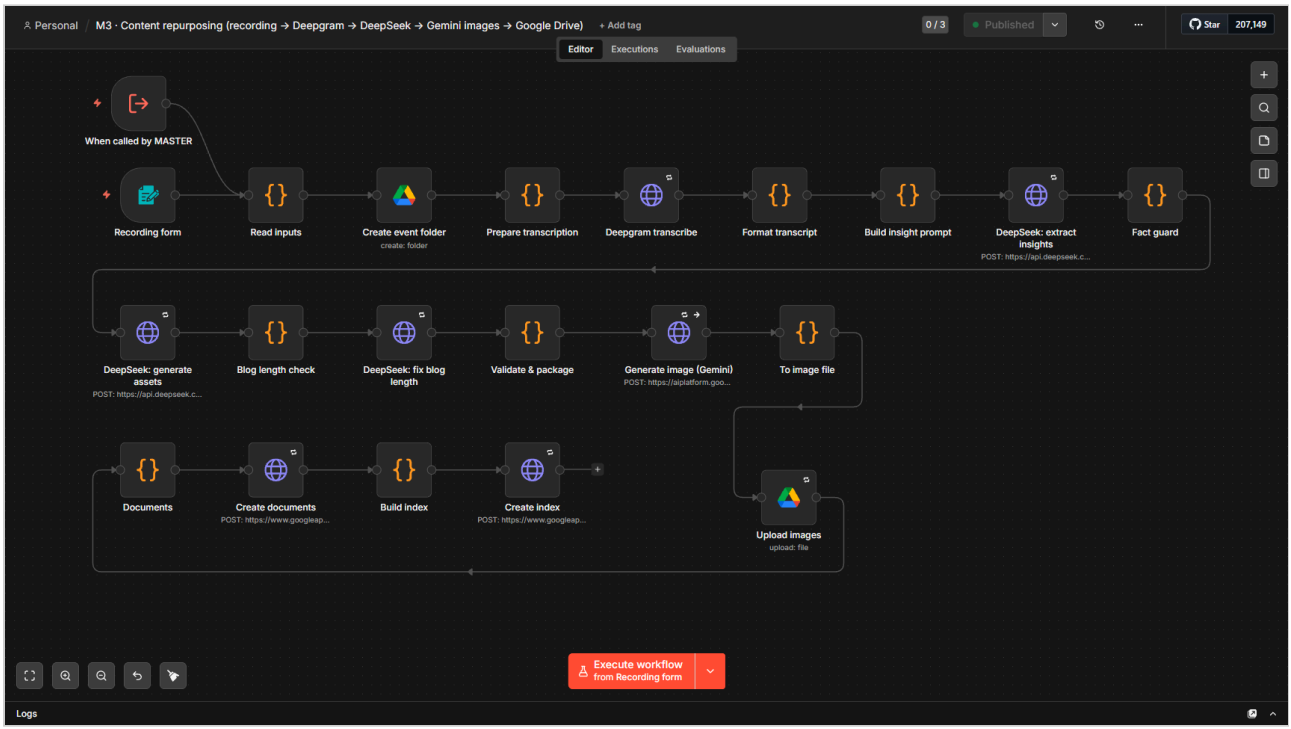
Blog draft as a formatted Google Doc.



Hub · Content package: latest package with image previews.



Documents and the automated quality checks.

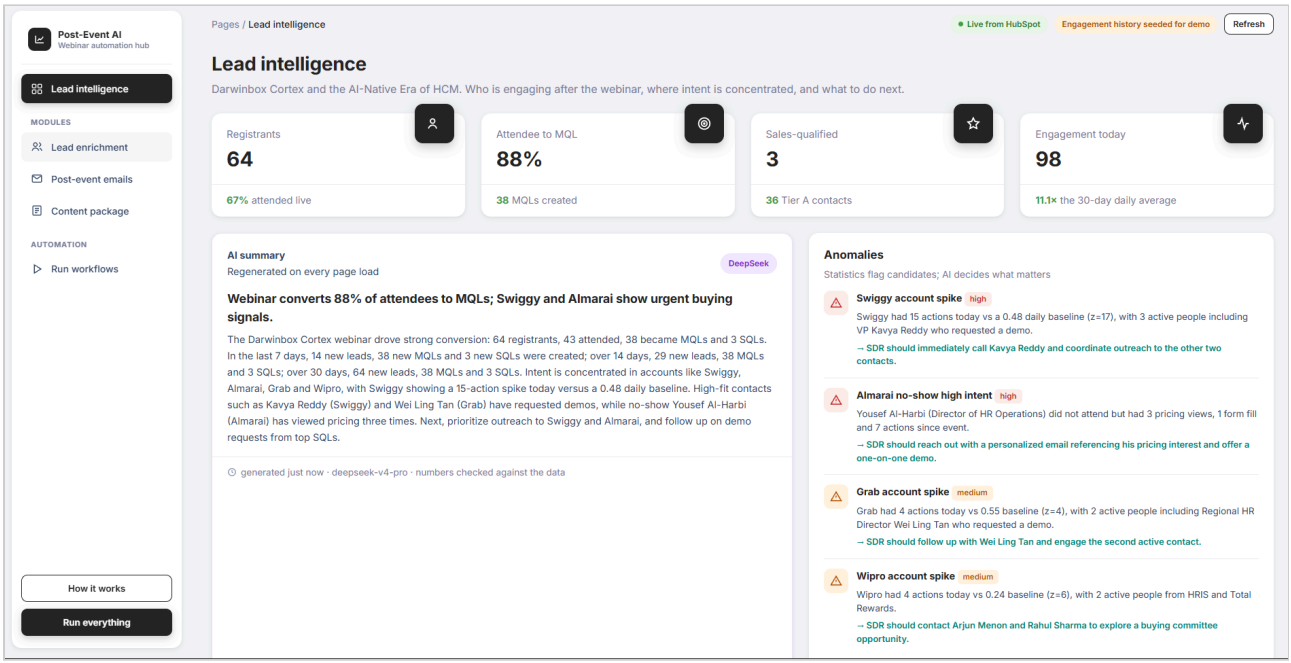


n8n · M3 workflow.

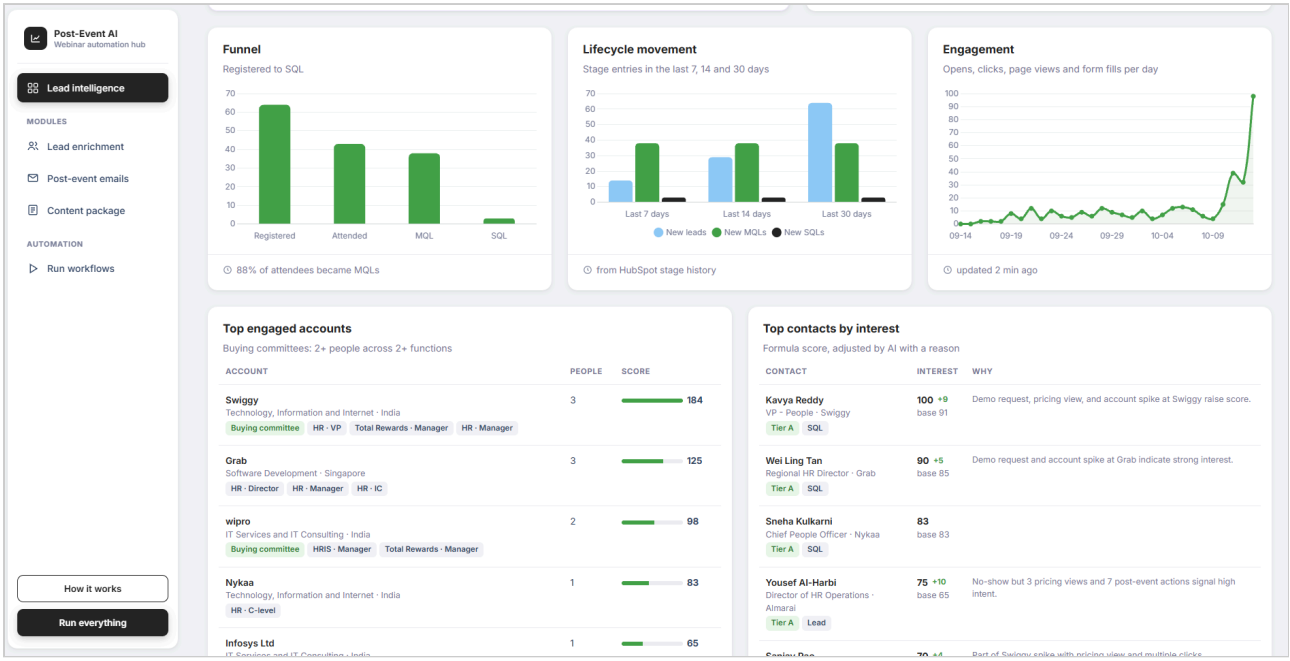
6. Module 4: Lead intelligence dashboard

A live page that reads HubSpot on every load. Code computes the metrics and statistical anomaly candidates; DeepSeek judges which anomalies matter, adjusts interest scores with a reason, and writes the narrative. A number guard checks that every number in the narrative exists in the data.

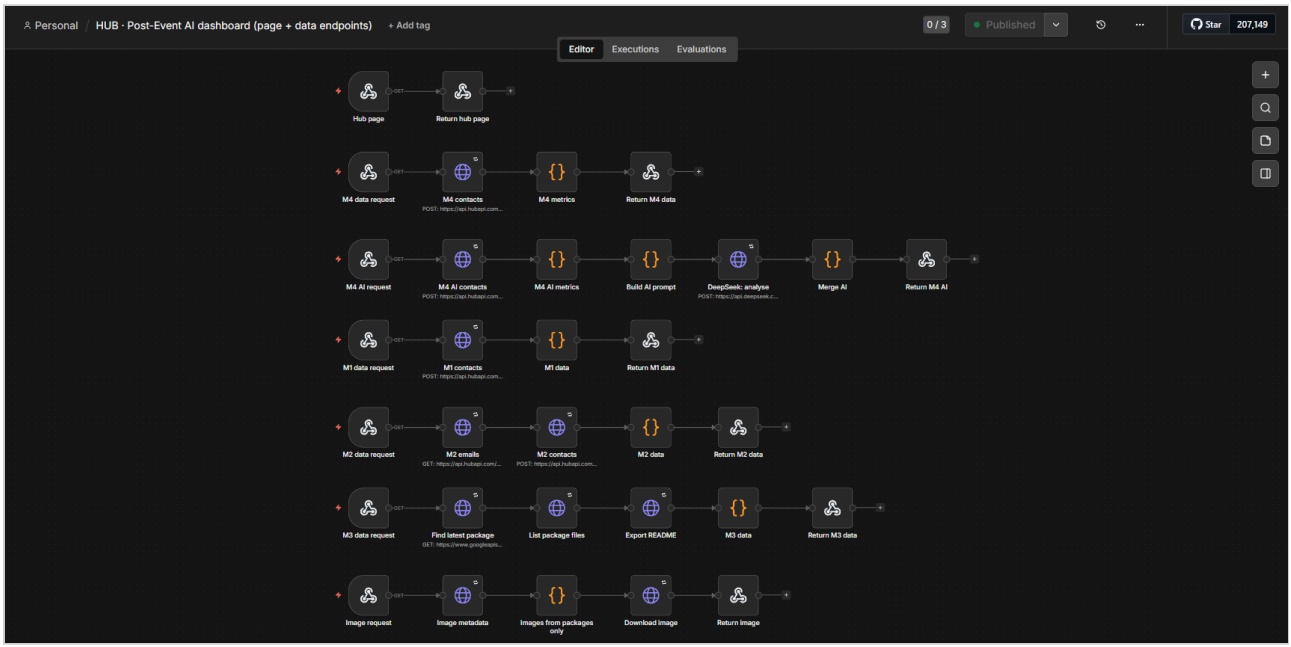
Brief asks for	What the build does
Attendee-to-MQL conversion	KPI cards and a Registered → Attended → MQL → SQL funnel (88% of attendees became MQLs; 3 SQLs from demo requests).
Top engaged accounts and buying committees	Accounts ranked by combined interest; a buying committee is 2+ people across 2+ functions, with each member's role and score.
Score lead interest	Formula base (attendance, minutes, questions, opens, clicks, page views, form fills, pricing views, recency decay) plus an AI adjustment of up to ±15 with a one-line reason.
Detect anomalies	Code flags candidates (per-account z-score spikes, silent Tier-A MQLs, no-shows with high intent); AI keeps what matters, removes duplicates and gives each one an SDR action. It found all three planted patterns.
Stage movement over 7, 14, 30 days; AI narrative refreshed on load	Lifecycle stage entries per window from HubSpot stage history; the narrative is regenerated on every page load.



Hub · Lead intelligence: KPIs, AI summary and anomalies with SDR actions.



Funnel, lifecycle movement (7/14/30 days), daily engagement, top accounts with buying committees, top contacts with AI-adjusted scores.

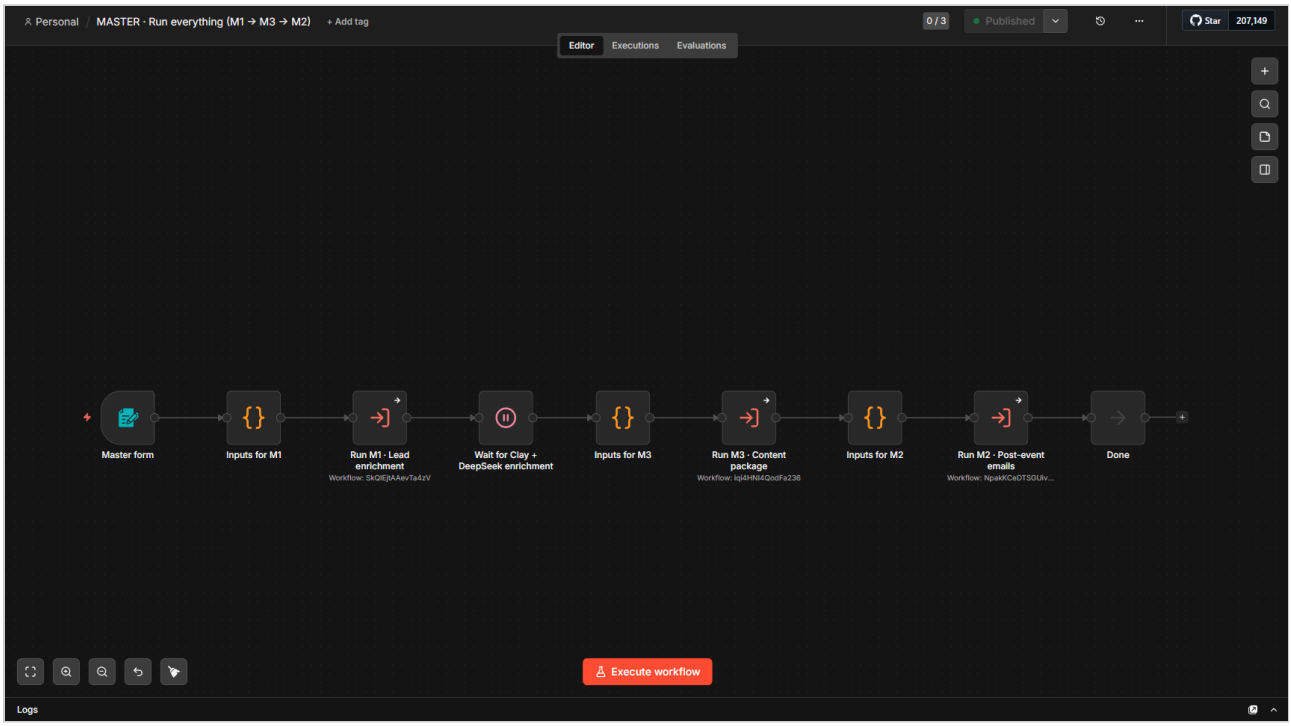


n8n · Hub workflow: the page plus one data endpoint per module and a guarded image proxy (only content-package images can be served).

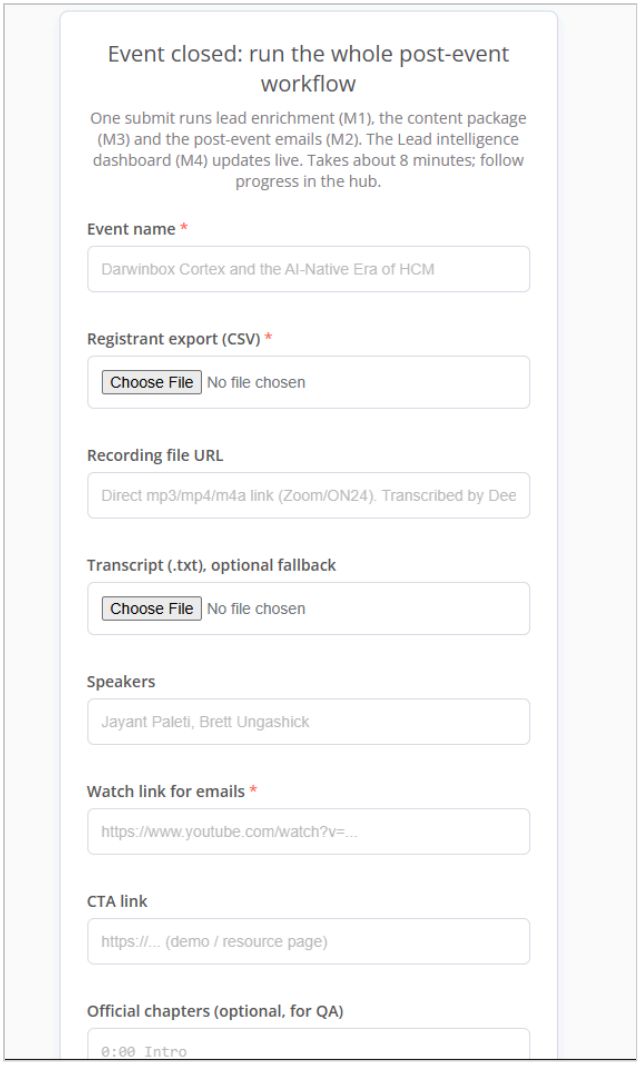
7. Control hub and "Run everything"

The hub is a single page served by n8n. It has a page per module, read live from HubSpot and Drive, and a "Run workflows" page linking every form. API tokens stay inside n8n.

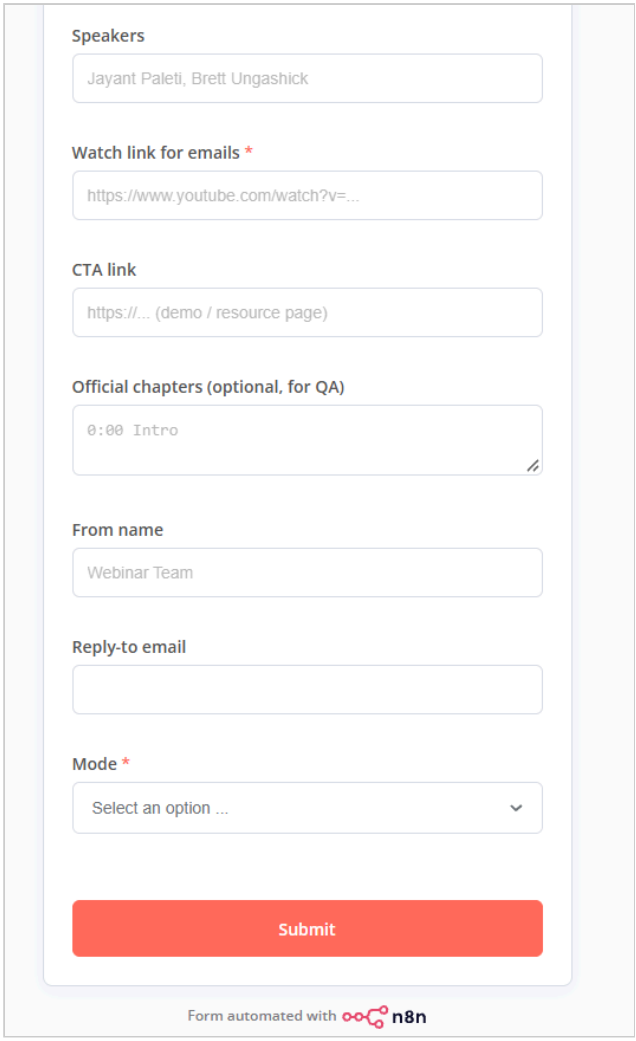
The master workflow takes everything in one form (registrant CSV, recording link or transcript, links, mode) and calls the modules in order: M1, then a wait while Clay and DeepSeek enrich every contact, then M3, then M2. Each module also has a "When called by MASTER" trigger next to its form, so the modules stay independent. The modules are individually tested end to end; the master is built and wired, and was not re-run for the submission to avoid duplicate sends.



n8n · Master workflow: M1 → wait for enrichment → M3 → M2.



Master form (top).



Master form (bottom).

8. AI guardrails and failure handling

Guardrail	Where	What it prevents
Fact guard	M2, M3	Every quote, evidence line and data point must appear word-for-word in the transcript; anything else is dropped and listed in the QA report. Emails are refused if fewer than 3 takeaways pass.
Number guards	M2, M3, M4	Numbers in emails, blog, infographic and narrative must exist in the transcript or the data.
Rules enforced in code	M3	Blog word range, meta length, YouTube chapter rules, X 280 characters, distinct hooks, 5-7 infographic sections.
Computed, not generated	M1, M2, M4	ICP points, completeness, speaker snapshot, funnel, stage movement and anomaly statistics are code; AI adds judgement and language.
Transparent inference	M1	Every AI-inferred value is stored with a confidence and basis; the AI is told not to invent people or facts.
Tolerant JSON handling	All	Repairs common LLM JSON slips (fences, line breaks, stray quotes, bare hashtags, trailing commas); one failed asset degrades gracefully instead of stopping a run.
Retries and throttling	All	HTTP and Drive retries; Clay callbacks rate-limited for HubSpot; image calls spaced for quota.
Safe sending	M2	Drafts-only mode; demo contacts on reserved domains; forward-only lifecycle changes in M1.
Secrets	All	Tokens live in n8n credentials only; the browser never sees them; the Clay callback uses a secret path; the image proxy serves only content-package images.

9. Testing, costs and next steps

How to see it live

- I can run the whole build live over a call: the hub, each module form, and the results in HubSpot, Clay and Google Drive.
- Every module runs from its own form or from the master form; M2 and the master have a **Create drafts only** mode so a demo sends no email.
- Sample inputs are in the repository (`data/registrants_zoom_export.csv`; transcript via the recording link).

Run cost

A full run costs a few rupees of LLM usage (DeepSeek) plus about three image generations; Clay, Deepgram, HubSpot, n8n and Drive run on free tiers or trial credit at this volume.

Next steps for production

- Trigger from the webinar platform (Zoom "recording completed" and registrant APIs) instead of forms.
- Paid Marketing Hub so every segment email is delivered; HubSpot workflows for follow-up sequences on the MQLs.
- Replace seeded engagement with HubSpot email events and page analytics; schedule the dashboard sync.
- A human approval step before sending (n8n "send and wait"), and per-brand style settings for images.